SUPERNEXA: Geometry-Aware Facial Expression Retargeting for Real-Time Animation

Abstract

Recent developments in computer vision and deep learning models have greatly improved the automated facial systems. Despite the enhancement the present facial retargeting technique, it still faces challenges with characters that have complex facial geometry, struggling to preserve realistic expressions throughout. Current retargeting techniques majorly rely on predefined mesh structures, which often leads to inaccuracy in expressions and motion stability. To overcome these challenges, the proposed study introduces Scalable Unified Performance Retargeting via Neural EXpressive Animation (SUPERNEXA), an efficient real-time architecture for facial expression retargeting. The proposed study integrates Thin Plate Spline(TPS) deformation, landmark based geometric modelling, gaze retargeting and GRU based temporal refinement into a unified framework. The proposed model is validated using several evaluation metrics. Experimental results demonstrates that the proposed architecture is scalable and efficient for high quality facial retargeting across different virtual characters.

1. Introduction

In the last decade, the field of computer vision has witnessed rapid growth in research, which is a result of the availability of advanced processors and models to analyze and interpret digital content. This growth has resulted in the development of effective computerized systems to work on visual data, creating realistic animated characters that can convey human like emotions through facial expressions and body movements. Demand for animation in movies, video games, and creation of virtual environment has increased due to widespread access to devices

While designing any animated character, the most significant part of the design is the face which conveys its emotions, personality and intentions. Audience engagement is highly influenced by performer's facial expressions, as they plays a crucial part in storytelling, helping the audience connect with the character. Many techniques have been developed previously for realistic facial animation. Traditional computerised models are either semi-automatic or majorly rely on manual intervention. Techniques like marker based motion capture, sensor capture, keyframe animation, blendshape editing are popularly used as part of the pipeline while creating animations[1,2]. These techniques produce professional results but they require specialized equipment, manual efforts and skilled pre and post processing[3]. Motion capture of characters requires multiple camera angels, dedicated studio setups, heavy processing, which results in high cost and complexity[4]. Manual intervention and expensive infrastructure limit the deployment of real-time interactive character generation[5].

Recent advancements in the field of animation are based on deep learning architectures, which propose automated tools for facial animation tasks such as facial motion extraction, landmark detection, and expression recognition. Modern architectures can analyze video and image frames to capture and track facial expressions without specialized tracking systems. Scanning through large datasets, these models can detect even the mildest variations in facial movements and facial features, thus extracting expression dynamics more precisely than traditional methods. AI driven facial animation models have been evolving, resulting in the creation of improved animation workflows without compromising on time, cost and effort[3].

Facial retargeting is an important part of automated deep learning models, working with expressions. Facial retargeting is used to replicate the facial movements of the human performer onto the face of an animated character, conveying human emotions and feelings through characters[2]. The most challenging part of this process is the difference in facial geometry between the human performer and the target character. Ineffective motion transfer can lead to fake expressions and distortion of face geometry, making retargeting an essential element in the pipeline for emotion transfer models[5].

Multiple models have been proposed in the past, for facial motion transfer from human to animated character. These proposed models do suffer from notable limitations, many of the existing solutions depend on predefined rigs or templated models that are restricted to specific mesh structures[1,2]. Such dependency restricts their diversity while working with arbitrary character models. Landmark-based motion capture methods have limitations in terms of the number of key facial points. Thus, at times failing to capture fine deformation details[3,6]. Loss of subtle motion penalizes the realism and expressiveness of the retargeted animations[5]. These challenges make it hard to preserve both expressive motion and identity characteristics, during the process of facial transfer.

This study proposes a novel method that achieves effective facial expression retargeting from a human performer to a digital character. The proposed SUPERNEXA(Scalable Unified Performance Retargeting via Neural EXpressive Animation) model architecture integrates motion extraction techniques with geometric deformation analysis, which accurately captures facial movements and effectively transfers them to the selected character even when having difference in facial geometry. The proposed architecture detects facial landmarks and constructs a geometric representation of the performer's face. Facial motions are further extracted by analyzing patterns based on displacement and deformation across facial regions. Further, the extracted motion is transferred to the animated face, while preserving the geometry of the target character.

2. Literature Review

The development of facial animation is majorly enhanced by the incorporation of recent technologies that allow a better representation of the underlying structure and dynamic behaviour of the face. This section provides an overview of existing research on facial landmark detection, motion extraction, and motion retargeting methods.

2.1 Facial Landmark Detection

Facial landmark detection has evolved from conventional handcrafted features and regression/model trained approaches to customised deep learning-based systems. Earlier researches frequently used Convolutional Neural Networks (CNNs) for facial landmark detection, ensemble regression trees were employed to identify facial keypoints. These techniques are ineffective at accurately estimating the same landmarks in other poses with alternate lighting and numerous occlusions[7][8]. Heatmap-based CNN architectures have enhanced the performance of landmark localization by using a model to predict a probability map of all of the landmarks thus accurately localizing facial structures[35]**.** CNN architectures struggle to handle extreme head poses and deal with partial occlusions making it necessary to have transformer-based architectures to model the long-range dependencies between facial parts [9]. High-resolution network designs have been proposed to maintain spatial information at multiple levels throughout the network and to improve landmark localization stability[10]. Attention mechanisms has improved landmark detection within complex environments by focusing on important areas of the face such as the mouth, eyebrows and eyes [11][12].

Self-supervised and semi-supervised approaches leverage unlabeled data to improve landmark robustness [13][14]. In order to enhance the representation of facial pose and expressions, there are 3D landmark estimation frameworks that go beyond traditional 2D detection by combining attention-based deep learning with 3D Morphable Models (3DMM) [15][16]. Recent work also explores NeRF-based representations for dense facial modelling [17]. In order to guarantee smooth facial motion tracking, temporal consistency constraints have also been integrated across video frames [20]. First-Order Motion Model techniques employ unsupervised keypoint detection for facial movement capture and are not reliant on pre-placed anatomical landmarks [18], [19].

2.2 Motion Extraction in Facial Animation

Deep learning motion extraction models have significantly contributed to the development of facial dynamics modeling, using input from video, audio, and motion to create dynamic models of facial motion. Earlier methods of estimating motion fields between frames used optical flow methods or parametric face models had limited ability to capture complex, non-rigid facial deformations[18], [23]. To effectively capture non-rigid deformations and maintain identity during animation, later advancements included occlusion maps and dense motion fields [19]. Disentangled motion representations were created to allow detailed control of facial dynamics (gaze, emotions) by merging head poses, lip movements, and facial expressions into separate latent components [24]. Few researches focused on extracting motion data from audio sources by utilizing speech audio to generate facial motion parameters that can produce synchronized lip motions, rhythmic head motion, head syncronization, and emotional expressions [21,22,26]. A systematic analysis of audio-based facial animations has demonstrated that while these deep learning platforms can generate precise small-scale motion, they may fail to produce temporally consistent images and smooth transitions among different individuals that were not previously included in the model's training set[20].

2.3 Motion Retargeting Techniques

Facial reenactment and avatar animation depend upon motion retargeting, where motion is transferred from a source to a target while retaining the target identity. Early techniques were based on three dimensional morphable models (3DMMs), which applies motions to target face using motion parameters from the source video[37]. However, due to simplified geometric and texture representations these methods often produce low-quality or low-resolution outputs[30]. Deep generative models have focused on enabling identity-preserving facial animation. Generative Adversarial Networks (GANs) have been widely used to create realistic talking faces by conditioning image generation on motion representations. Systems like Face2Face and subsequent GAN-based approaches demonstrated effective transfer of facial expressions and head motion between identities[20]. To improve the separation of identity and motion, later techniques introduced enhanced motion normalization and appearance adaptation modules[19],[31]. Disentangled representation learning has further improved retargeting quality. MakeItTalk proposed a speaker-aware disentanglement technique that separates lip motion, gaze direction, and head pose into independent components [24]. NeRF-based approaches have further enhanced visual realism by reconstructing high-fidelity 3D facial representations and applying motion signals[33]. Additionally, controllable speaking style transfer allows expressive animation while preserving identity[34]. Motion retargeting has also been extended to full-body animation frameworks, enabling temporally consistent transfer across identities [32].

2.4 Research Gaps and Contributions

Researchers have proposed efficient proposals for facial animation and retargeting still the proposed models suffer from fundamental limitations. Existing methods heavily relies on predefined templates, fails at accurate depth aware motion representation, computational complexity, incomplete gaze modelling and fails at adaptive temporal consistency. These limitations collectively impact the realism, scalability, and robustness of facial retargeting systems, particularly when applied to diverse and unseen character geometries. This study aims to propose a novel architecture named SUPERNEXA which aims to overcome the identified gaps when applied to diverse and unseen characters. Table 1, summaries the identified research gaps(G1-G5) and the proposed contribution(PC1- PC5) which forms the foundation of SUPERNEXA architecture with the aim to overcome identified gaps.

Table 1. Gap-to-contribution mapping

Gaps in Prior Art (G 1 to 5)	**Proposed Contribution (PC 1 to 5)**
G1: Landmark-based methods operate in the image plane and fail to capture depth-direction micro-expressions[2],[32].	PC1: Normal-flow augmentation: recovers depth-direction expression signals using surface-normal projected optical flow.
G2: Existing methods depend on predefined rigs, templates, or manual region annotation, limiting generalization[1].	PC2: Curvature-based region detection: automatically assigns semantic facial regions without templates or training data.
G3: Conventional TPS implementations solve redundant per-axis systems, increasing computational overhead[28][36].	PC3: Batched multi-axis TPS (GPU): eliminates redundant computations using a unified matrix solve, improving efficiency.
G4- Gaze retargeting methods model only lateral iris motion, ignoring depth recession and causing unrealistic artefacts[32],[39]	PC4: Spherical-recession gaze model: incorporates depth-aware iris motion using geometric eye modelling.

G5- Fixed-parameter temporal filters cannot adapt to performer-specific dynamics and over-smooth rapid expressions[2],[17],[38].	C5: Online GRU temporal prior: learns adaptive, performer-specific temporal dynamics for stable expression tracking.

3. Methodology

The proposed SUPERNEXA architecture addresses the challenge of transferring facial expressions across characters with varying geometries and topologies. Differences in shape, scale, and structure often lead to distorted or unrealistic expression transfer. To mitigate this, the framework employs a geometry-aware pipeline designed to preserve expression dynamics and semantic consistency. As shown in Fig. 1, the proposed SUPERNEXA architecture consists of three stages namely (i) landmark detection and geometric representation, (ii) motion extraction, and (iii) geometry-aware retargeting, which together enables robust and consistent facial expression transfer.

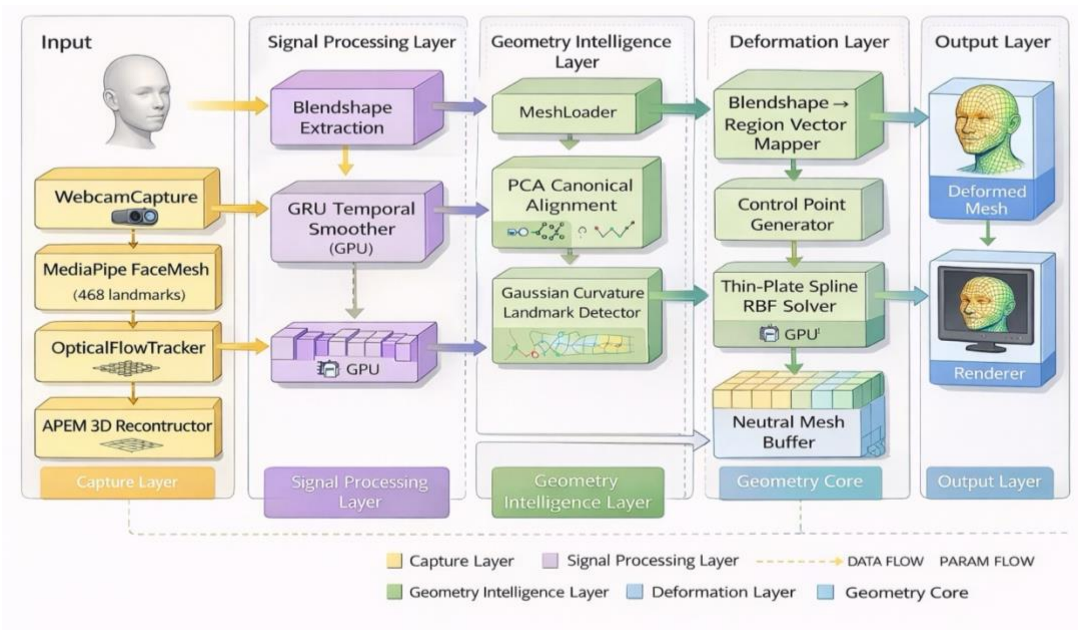

Figure 1. Overall architecture of the proposed SUPERNEXA framework for facial expression retargeting across heterogeneous facial geometries

3.1 Dataset

The testing dataset used for evaluating the proposed SUPERNEXA framework consists of four human performers and four heterogeneous target character models. The performer subset includes two male and two female subjects, with 64 RGB facial images captured for each performer at a resolution of 512×512 pixels under varying expressions and gaze movements. The dataset is designed focusing on evaluation of cross-topology generalization of the proposed SUPERNEXA architecture, rather than large-scale training performance. The target characters include a Na'vi-inspired humanoid, a Toruk-inspired fantasy creature, a tiger-based non-human model, and a stylized cyberpunk female character. These characters were selected due to their significant differences in facial topology and geometry, enabling evaluation of the framework's ability to preserve facial expressions across highly dissimilar target structures. Additionally, the framework was initialized using a pretrained facial expression dataset of 1000 images obtained from a publicly available Hugging Face repository for learning generalized facial motion and expression dynamics[40].

3.2 Stages of the proposed SUPERNEXA Architecture

The SUPERNEXA stages are combination of geometric representation, motion modelling and deformation mechanisms. Fig. 2 shows the flowchart demonstrating the mathematical foundation of conversion of input facial features to retargeted mesh output using proposed SUPERNEXA architecture. In Stage 1, facial landmarks are detected from the input image using a FaceMesh model, followed by stabilization and geometric representation to obtain accurate feature points. Stage 2 focuses on motion extraction, where landmark displacements are computed, refined using anatomical constraints, and decomposed into blendshapes with additional motion enhancement. Finally, Stage 3 performs geometry- aware retargeting by selecting control points and applying deformation techniques to generate the final output mesh, where the facial motion is successfully transferred to the target image.

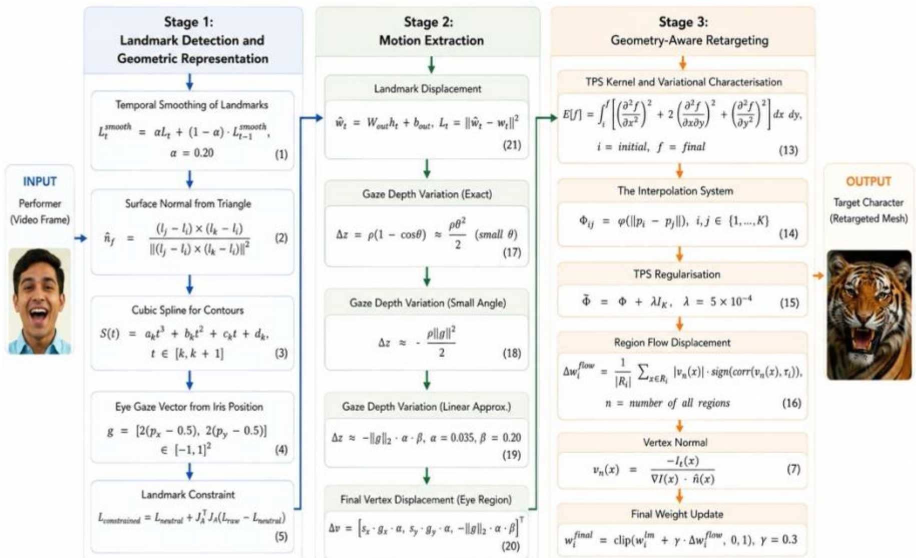

Figure 2. Flowchart of the proposed SUPERNEXA framework illustrating the algorithmic workflow for real-time facial expression retargeting.

3.2.1 Stage 1- Landmark Detection and Geometric Representation

The first stage extracts 3D facial landmarks representing key facial regions. To reduce noise and temporal instability, an exponential moving average (EMA) filter is applied, as shown in Eq. (1). The stabilized landmarks are connected using a predefined triangulation to form a facial mesh. Surface normal vectors are computed from neighbouring triangular faces using the cross product formula as shown in Eq. (2), which represents the local surface direction required for later deformation and retargeting stages. To represent smooth facial structures such as the lip and eye movements, ordered ladnmark points are fitted using cubic spline curves as defined in Eq. (3). Additionally, iris landmarks are used to compute a normalized gaze vector (Eq. 4), with separate smoothing to preserve natural eye motion. These features collectively form the geometric representation used in subsequent stages.

$$L_t^{smooth} = \alpha\, L_t + (1 - \alpha) \cdot L_{t-1}^{smooth}, \quad \alpha = 0.20 \tag{1}$$

$$\hat{n}_f = \frac{(l_j - l_i) \times (l_k - l_i)}{\|(l_j - l_i) \times (l_k - l_i)\|2} \tag{2}$$

$$S(t) = a_k t^3 + b_k t^2 + c_k t + d_k, \quad t \in [k, k+1] \tag{3}$$

$$g = [2(p_x - 0.5),\ 2(p_y - 0.5)] \in [-1, 1]^2 \tag{4}$$

3.2.2 Stage 2- Motion Extraction

The second stage uses blendshape parameters to convert the geometric feature into a compact motion representation. The facial motion is represented by weighted blendshape activations, not by using landmark coordinates directly. First, the displacement of the landmarks from a neutral position is computed, and then anatomical constraint projection Eq. (5) is applied to ensure physically plausible motion . Then the constrained motion is decomposed into blendshape weights with least squares optimisation. The resulting sparse blenshape vector acts as the motion descriptor for retargetting stage. At last these deformations mapping are used to assign blendhape weights to different facial regions and corresponding control points of mesh.

$$L_{constrained} = L_{neutral} + {J_A}^{\dagger} JA(L_{raw} - L_{neutral}) \quad (5)$$

3.2.3 Stage 3- Retargeting

The retargeting stage is the core component of the proposed SUPERNEXA architecture. Retargeting applies the extracted facial motion to a target mesh that may differ in structure and geometry. Given the blendshape weights and gaze vector for frame 't', the retargeting module computes a deformation field that converts the neutral mesh into the expression dependent configuration as shown in the algorithm mentioned in Table 2. To achieve this, the retargeting module integrates depth-aware motion enhancement, automatic region detection, deformation propagation, and gaze modelling into a unified pipeline.

3.2.3.1 Normal-Flow Blendshape Augmentation

The proposed architecture incorporates a normal-flow-based augmentation module to overcomes the limitations of existing facial motion beyond landmark-based limitations. This approach allows recovery of depth-direction facial deformations not captured by conventional image-plane analysis, in three sequential steps explained further in this section.

Dense Optical Flow. Blendshape estimation based on landmarks primarily captures motion in the image plane, and thus, does not fully represent deformation occurring along the facial surface along the depth axis. Minute facial details, such as cheeks bulging, nostril widening, and tightening of lips, play a crucial for realistic expressions synthesis, but these details cannot be captured by landmark movement alone. To capture these fine movements, dense optical flow is computed between consecutive frames using the Farneback polynomial expansion method. This approach assumes that pixel brightness remains relatively the same between frames and uses the intensity variation of the image over time and space to estimate the pixel movement, as described in Eq. (6), where Ix, Iy are the spatial image gradients, and It is the temporal gradient. The Farneback method resolves the aperture problem (the inability to determine flow direction from local intensity change alone) by propagating constraint information across a Gaussian pyramid, producing reliable flow at feature-rich and feature- sparse regions alike.

$$I_x(x)\, u(x) + I_y(x)\, v(x) + I_t(x) = 0 \quad (6)$$

Normal-Flow Projection. Two-dimensional optical flow is mapped onto surface normal vectors of the reconstructed facial mesh, as defined in Eq. (7). This stage extracts motion in the outward direction from the facial surface. The resulting scalar quantity, is known as normal flow, it represents motion along the direction of the facial surface. The deformation along the facial surface direction, is captured by normal flow. This provides additional information that landmark dependent motion cannot capture.

$$v_n(x) = \frac{-I_t(x)}{\nabla I(x) \cdot \hat{n}(x)} \quad (7)$$

Aggregation and Fusion. Normal flows are integrated for each vector to compute additional deformations of the facial mesh, as expressed in Eq. (8). This integration estimates the motion direction using predefined deformation patterns, ensuring that the detected displacement movement matches the facial muscle movement. The resulting contribution, is combined with landmark-based blendshape weights through a bounded weighted integration ensuring numerical stability and improving expression quality as formulated in Eq. (9).

$$\Delta w_i^{flow} = \frac{1}{|R_i|} \sum_{x \in R_i}^{n} |v_n(x)| \cdot sign\big(corr(v_n(x),\ \tau_i)\big), n = number\ of\ all\ regions \quad (8)$$

$$w_i^{final} = clip\big(w_i^{lm} + \gamma \cdot \Delta w_i^{flow},\ 0,\ 1\big),\ \gamma = 0.3 \quad (9)$$

3.2.3.2 Curvature-Weighted Automatic Region Detection

To achieve topology-independent retargeting, a curvature-weighted method consisting of 3 stages is used to automatically detect semantically meaningful facial regions based on mesh geometry.

Stage 1 — PCA Orientation Normalisation

Since the character mesh can be placed in any orientation, the first stage aligns the target mesh into a canonical coordinate frame using Principal Component Analysis (PCA). PCA analyses the covariance of vertex positions by calculating the principal directions of the mesh and represents the major spatial directions of the mesh, as shown in Eq. (10). This orientation normalisation ensures consistent spatial interpretation of facial geometry prior to region identification.

$$[e_1, e_2, e_3] = PCA(V),\ V \in \mathbb{R}^{N \times 3} \quad (10)$$

Stage 2 — Gaussian Curvature by the Angle-Deficit Method

To identify important facial features, Gaussian curvature is used to compute a saliency value for each vertex. The angle- deficit formulation is applied to estimate curvature, which further calculates the variation of angle around a vertex from its expected value on a flat surface, as defined in Eq. (11). Vertices with high curvature usually corresponds to crucial boundaries of facial geometry, such as eye rims, lip contours, and nostril edges. These vertices are well suited for use as deformation control points

$$\kappa_G(v_i) = 2\pi - \sum_{i \ni v_i}^{f} \theta_f(v_i);\ i = initial\ , f = final \quad (11)$$

Stage 3 — Face Isolation, Projection, and Region Assignment

Vertices corresponding to in facial regions are identified based on the position and direction of the surface, removing non- facial region. The remaining vertices are assigned to a normalized facial coordinate system, ensuring consistent placement of the face in a standard space independent of its size, as expressed in Eq. (12). Within this space, specific regions depict different sections of the face. Within these areas, vertices with the highest curvature are selected as control points for deformation.

$$\tilde{x}_i = \frac{x_i - x_{min}}{x_{max} - x_{min}},\ \tilde{y}_i = \frac{y_i - y_{min}}{y_{max} - y_{min}} \quad (12)$$

3.2.3.3 Batched Multi-Axis Thin-Plate Spline Deformation on GPU

This subsection introduces a TPS-based deformation model for smooth propagation of control point displacements, along with a batched formulation for efficient multi-axis computation.

TPS Kernel and Variational Characterisation : For the propagation of the control points displacement across the mesh surface, the framework utilizes the Thin-Plate Spline (TPS) deformation model. TPS provides smooth interpolation of scattered deformation constraints using a radial basis kernel defined as a function of Euclidean distance. The deformation field minimizes the bending energy functional, resulting in smooth surface after the deformation is applied, as expressed in Eq. (13).

$$E[f] \;=\; \iint_i^f \left[\left(\frac{\partial^2 f}{\partial x^2}\right)^2 \;+\; 2\left(\frac{\partial^2 f}{\partial x\,\partial y}\right)^2 \;+\; \left(\frac{\partial^2 f}{\partial y^2}\right)^2\right] dx\,dy, i = initial\,, f = final \tag{13}$$

The Interpolation System: The relationships between the control points are computationally depicted with a Gram matrix containing pairwise kernel evaluations, as formulated in Eq. (14). This matrix depicts how displacement from one control point propagates to other control points during the deformation process. To improve numerical stability, a small regularisation term is added to the diagonal of the system matrix as shown in Eq. (15).

$$\Phi_{ij} \;=\; \varphi\left(\,\|p_i \;-\; p_j\,\|\right),\;\; i,j \;\in\; \{1,\dots,K\} \tag{14}$$

$$\widetilde{\Phi} \;=\; \Phi \;+\; \lambda\, I_K,\;\; \lambda \;=\; 5 \;\times\; 10^{-4} \tag{15}$$

Novel Batched Multi-Axis Formulation: The deformation issue is computed for all spatial axes concurrently, using a single matrix of displacement vectors, instead of solving each axis independently. The precomputed evaluation matrix applies the deformation weights to all vertices of the mesh, creating the final displacement field. The modified vertex configuration is gained by adding the deformation offsets to the mesh coordinates.

3.2.3.4 Iris Gaze Retargeting with Spherical-Recession Geometry

To improve realism in facial animation, a spherical geometry-based gaze retargeting model is introduced to capture both lateral and depth-aware iris motion.

Gaze Vector Extraction. For realistic facial expression, the gaze vector must be accurately reproduced. The gaze direction is evaluated using the iris centroid with respect to the eye coordinate frame, resulting in a normalized 2D gaze vector, as shown in Eq. (16).

$$g \;=\; \left[2(p_x \;-\; 0.5),\; 2\left(p_y \;-\; 0.5\right)\right] \;\in\; [-1,1]^2 \tag{16}$$

Derivation of the Depth Recession. Assuming the eyeball to be spherical in shape, the displacement of the iris center during the gaze changes can be evaluated using spherical rotation geometry. This describes how the gaze angle varies as the iris moves back from its neutral position, as defined in Eq. (17). For a small angular deviation, the Taylor expansion is used to simplify the equation, resulting in a simpler quadratic equation, as expressed in Eq. (18). For a very small gaze magnitude, a linear approximation is applied to ensure numerical stability, as given in Eq. (19).

$$\Delta z \;=\; \rho(1 \;-\; \cos\theta) \;\;\approx\;\; \frac{\rho\theta^2}{2} \;\; (small\,\theta) \tag{17}$$

$$\Delta z \approx -\frac{\rho\, |g|^2}{2} \quad (18)$$

$$\Delta z \approx -\|g\|_2 \cdot \alpha \cdot \beta,\ \alpha = 0.035,\ \beta = 0.20 \quad (19)$$

Final Vertex Displacement The final displacement of the eye region is derived by integrating lateral translation from the gaze vector with the calculated depth recession, as formulated in Eq.(20). This method generates realistic iris motion similar to spherical eye rotation, without requiring a rigged eyeball model.

$$\Delta v = \left[s_x \cdot g_x \cdot \alpha,\ s_y \cdot g_y \cdot \alpha,\ -\|g\|_2 \cdot \alpha \cdot \beta\right]^T \quad (20)$$

Table 2: Algorithm for Mesh Sequence Processing

Algorithm 1: Mesh Sequence Processing Procedure

Require: Source mesh sequence {Vs(t)}, Neutral mesh Vref, Target mesh Vt, Basis B, Control points C
Ensure: Output transformed mesh sequence
Precomputation: Align meshes using Procrustes / ICP
$P \leftarrow (B^T B + \lambda I)^{-1} B^T$
Factorize $\Phi \rightarrow \Phi ij = \varphi(||Ci - Cj||),\ \varphi(r)=r^2 \log(r+\varepsilon)$
Initialize h(0)=0
for t = 1 to T do
    $\Delta V(t) = Vs(t) - Vref$
    $\delta(t) = vec(\Delta V(t))$
    $w(t) = P \cdot \delta(t)$
    $z(t) = \sigma(Wz\, w(t) + Uz\, h(t-1) + bz)$
    $\tilde{h}(t) = \tanh(Wh\, w(t) + Uh\, h(t-1) + bh)$
    $h(t) = (1 - z(t)) \odot h(t-1) + z(t) \odot \tilde{h}(t)$
    $D(t) = \Delta V(t)[C]$
    $W(t) = \Phi^{-1} D(t)$
    Compute ΦV
        $V't(t) = Vt + \Phi V\, W(t)$
end for
Output: {V't(t)}

3.3 Online GRU Temporal Prior for Blendshape Smoothing

The blendshape weight sequence obtained during expression estimation may contain frame-to-frame fluctuations caused by detection noise and motion estimation variability. To improve temporal consistency, the framework incorporates a recurrent model based on a single-layer gated recurrent unit (GRU). The GRU receives the blendshape weight vector as input and updates an internal hidden state that captures temporal dependencies between consecutive frames. The hidden representation is mapped to the predicted blendshape vector through a linear output transformation, as expressed in Eq. (21).

$$\hat{w}_t = W_{out} h_t + b_{out},\ \mathcal{L}_t = \|\hat{w}_t - w_t\|^2 \quad (21)$$

Online Per-Performer Adaptation. The GRU parameters are updated online during runtime instead of relying on offline training. Periodic gradient updates are performed over short temporal windows, enabling the network to adapt to the current performer's motion characteristics. This allows the model to capture performer specific behaviour, such as blinking rate, expression timings, and jaw motion.

Warm-Up Blending. During initial frames, the recurrent model lacks sufficient temporal context, which affects stable predictions. To address this limitation, a blendshape mechanism is used, which gradually combines the

GRU output with the original blendshape signal. As the model context increases, the GRU becomes a significantly important part while the landmark signal remains the primary reference, ensuring the stable and responsive facial expression tracking.

Why GRU over alternatives? Compared to the Kalman filter, which assumes linear motion and Gaussian noise, the Butterworth filter, which introduces a fixed phase lag proportional to its order, and an Exponential moving average, which operates on a single frequency parameter, the online GRU learns a nonlinear model of performer-specific blendshape dynamics in real time. After roughly 450 frames at 15s at 30FPS, the hidden state encodes a model of blink frequency, jaw motion, brow asymmetry, and smile curvature. The GRU functions as a predictive prior rather than a lag-inducing filter, anticipating the next state from current momentum and perceptually reducing temporal latency.

4. Experimental Results and Discussion

This section evaluates the proposed SUPERNEXA framework using quantitative measurements, ablation studies, and qualitative visualization analysis. The findings emphasize the role of the spatial and temporal modules, show robustness to different facial expressions and identities, and prove real-time capabilities on GPU platforms.

4.1 Evaluation Metrics

The validation study of the proposed SUPERNEXA architecture was performed with a tailored sequence of facial animation which consist of around 300 frames and five illustrative facial expressions, namely *jaw_open, smile, brow_raise, eye_blink, and cheek_puff*. The output characters used for facial retargeting were *Na'vi, Tiger, Toruk and Cyberpunk*. For the performance assessment of the proposed architecture and to rationalize the contribution of individual components of the pipeline, four configurations were analyzed. Vanilla pipeline without TPS deformation and temporal smoothing (C1), Vanilla pipeline with temporal smoothing (C2), Vanilla pipeline with TPS deformation (C3), and the proposed SUPERNEXA architecture with the combination of spatial deformation and temporal smoothing (C4). The result obtained using these configurations are examined using four essential evaluation measures namely, Landmark Error, Motion Jitter, Deformation Accuracy and Coverage. Landmark Error (LE) calculates the mean Euclidean distance between actual and predicted facial landmarks points. It determines the actual displacement in magnitude between the predicted blend-shape weights and the target performer for each expression on the mesh. Motion Jitter (MJ) is evaluating frame to frame divergence of the displaced points between two consecutive frames. This metric measures performance of temporal smoothing component as TPS deformation has no effect on motion fluctuation. Deformation Accuracy (DA) determines fidelity of expression transfer by computing Pearson correlation between predicted and target displacement sequences, averaged across all expressions, whereas Coverage quantifies the number of facial mesh vertices that are actively involved in displacement, indicating spatial extent of expression.

4.2 Ablation Study

The ablation study is used to examine the individual and combined impact of the TPS deformer and the GRU-based temporal smoother within the proposed architecture. This examination helps in understanding the contributions of the spatial and temporal components in improving the performance of the facial retargeting process. Table 3. shows performance comparison of different ablation settings across all evaluation metrics.

Table 3. Overall performance of four ablation configurations across four metrics

Configuration	LE	MJ	DA	Coverage
C1	0.0269	0.0736	0.9526	0.445
C2	0.0218	**0.0226**	**0.9853**	**0.467**
C3	0.0096	0.0736	0.9526	0.445
C4	**0.0078**	**0.0226**	**0.9853**	**0.467**

4.2.1 Effect of TPS deformation

Comparing configurations C1 and C3 from Table A highlights the need for and impact of the TPS deformer in facial retargeting process. LE is reduced from 0.0269 to 0.0096, resulting in a reduction of 64.3%, as the TPS deformer robustly locates the facial deformations in the correct local expression region, and spatial weighting is integrated based on geometric closeness of each region. Without TPS, however, there is leakage in the resulting deformation as the blendshape weights are applied uniformly over the mesh. TPS deformer neither affects temporal domain nor impacts the number of vertices as there is no change in ME, DA, and coverage metrics for C1 and C3. Thus, inclusion of TPS results in significant improvement in spatial accuracy.

4.2.2 Effect of temporal (GRU) smoothing

The impact of including the GRU-based smoother can be analyzed by contrasting the results achieved for C1 and C2 in Table 3. The MJ value has decreased from 0.0736 to 0.0226, illustrating a drop of 69.3%, which validates the capacity of the smoother to attenuate the high-frequency noise in the predicted expression sequences, resulting in a more stable facial motion without any trembling effect. There is a slight improvement in LE and DA, as smooth temporal predictions were able to follow the latent expression sequences resulting in enhancements in spatial alignment. The subtle improvement in coverage ensures a continuous activation of significant facial areas.

4.2.3 Synergistic Effect

The fusion effect of TPS deformer and GRU smoother included in pipeline C4 can be examined by juxtaposing the results obtained in Table 3 for configurations C1 and C4. Across all four metrics, C4 achieves optimal performance. LE has decreased from 0.0269 to 0.0078, showing a total reduction of 71.0%, while MJ has also dropped from 0.0736 to 0.0226, corresponding to a 69.3% improvement, similar to the pipeline with only the temporal smoother. The rise in DA from 0.9526 to 0.9853 shows a strong association due to the reduction in noise. The modest increase in coverage value from 0.445 to 0.467 shows consistent engagement of relevant facial regions. This study depicts that the combining TPS and GRU components is jointly beneficial for facial expression retargeting. While the inclusion of TPS yields spatial enhancement by precisely choosing the vertices to move, GRU offers temporal refinement by smoothing the motion of those vertices. The proposed hybrid SUPERNEXA architecture ensures that the amalgamation of both modules provides the ideal configuration without interference.

4.3 Per-Expression Performance Analysis

The significance of this subsection is to demonstrate the robustness and reliability of proposed architecture for any type of expression. This is crucial for facial retargeting because different expressions vary in terms of mesh displacement, complexity, and magnitude. The highest displacement expression, *jaw_open,* depicts the largest LE values across all configurations, suggesting that high-displacement expression is more complex and difficult to retarget, whereas lower displacement expressions have resulted in lower LE values, indicating ease in retargeting subtle expressions. From Table 4, it is evident that for all five expressions, C4 configuration achieves the least LE value, with a mean LE dropping from 0.0269 (C1) to 0.0078. This illustrates that fusing the spatial and temporal advancements produces consistent performance enhancements across all types of expression.

Table 4. Per-Expression LE of four ablation configurations

Expressions	Displacement	Configurations			
		C1	C2	C3	C4
jaw_open	0.509	0.0742	0.0608	0.0265	**0.0217**
Smile	0.140	0.0203	0.0156	0.0073	**0.0056**
brow_raise	0.120	0.0154	0.0131	0.0055	**0.0047**
eye_blink	0.070	0.0108	0.0085	0.0039	**0.0030**
cheek_puff	0.095	0.0135	0.0107	0.0048	**0.0038**
Mean	—	0.0269	0.0218	0.0096	**0.0078**

Expression displacement and motion jitter have a positive correlation. As shown in Table 5, expressions with higher displacement, such as jaw_open, have larger value of MJ since there is a lot of motion in a greater number of vertices. Contrarily, expressions with low displacement implicitly have a smaller value of MJ. The incorporation of the temporal smoother (C2 and C4) has significantly reduced the motion jitter of expressions, specifically for expressions with higher displacements, producing reliable and smooth expression transfer.

Table 5. Per-Expression MJ of four ablation configurations

Expressions	Configurations			
	C1	C2	C3	C4
jaw_open	0.0767	0.0243	0.0767	**0.0243**
smile	0.0704	0.0215	0.0704	**0.0215**
brow_raise	0.0700	0.0212	0.0700	**0.0212**
eye_blink	0.0780	0.0253	0.0780	**0.0253**
cheek_puff	0.0729	0.0209	0.0729	**0.0209**
Mean	**0.0736**	**0.0226**	**0.0736**	**0.0226**

4.4 Qualitative Visual Evaluation

To strengthen the prior quantitative analysis, this subsection focuses on providing a qualitative examination of the SUPERNEXA system proposed above. Though quantitative parameters allow for objective analysis of the architecture's performance, visual assessment is crucial for evaluating its perceptual realism, anatomical accuracy, and temporal consistency of the morphed facial expressions. Qualitative findings are evaluated from the perspective of various characters and a set of exemplary expressions in order to emphasize the contribution of individual modules of spatial deformation and temporal smoothing.

4.4.1 Cross-Configuration Visual Assessment

However, upon close observation of all expressions for all characters involved in fig. 3, it is evident that there exists a definite hierarchy in the quality of perception amongst all configurations. The configuration, C1, yields the least favorable outcome, wherein the deformations produced are rough, inaccurate, and even anatomically impossible in nature, suggesting a lack of spatial and temporal control. C2 and C3 do yield some improvement over C1 but the difference between the two is insignificant, visually speaking, as C2 yields smoother animation with less jitter compared to C3, which performs better spatially. However, both still exhibit problems such as mislocalization and a lack of flowiness that prevent them from yielding convincing expressions. C4, on the other hand, (or SUPERNEXA) yields expressions that are both accurately localized and temporally seamless. They exhibit smooth and natural deformations with stable animations that result in expressions that are realistically expressive. Thus, it becomes evident that only through the successful integration of spatial deformation and temporal smoothing will highly convincing results be achieved.

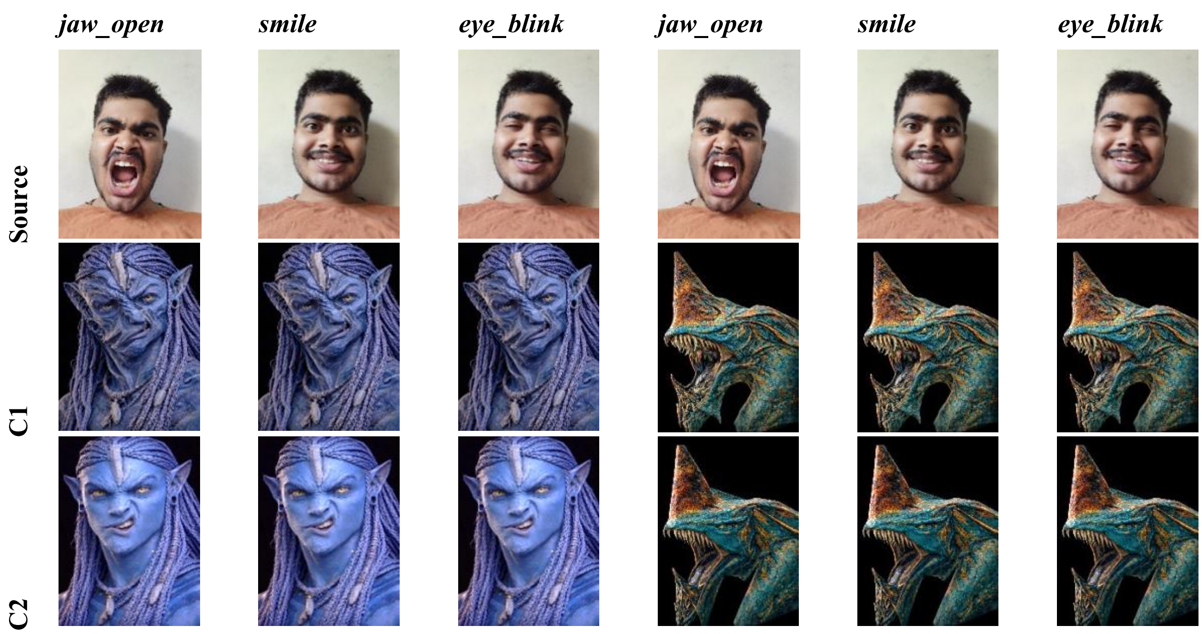

Fig 3. Qualitative perceptual comparison of the proposed SUPERNEXA architecture across multiple expressions (jaw open, smile, eye blink) and different configurations (C1–C4).

4.4.2 Landmark-Guided Facial Retargeting: Cross-Configuration Evaluation

The Fig. 4 shows how the facial landmarks guide the deformation process in each configuration (jaw_open, smile, and neutral). The facial landmarks are consistently detected and precisely model the geometry of the source face in each expression. Nevertheless, there is variation in how these landmarks affect the deformation process according to the particular configuration. It is also important to note that while the levels of expression differ, ranging from strong displacements like jaw_open to low displacements in neutral, the qualitative behavior for all four setups stays consistent with the findings from Fig. 3. The following holds true: C1 produces the lowest level of accuracy, C2 provides an improvement over time, C3 is a good improvement in space, while C4 always produces the highest anatomical consistency.

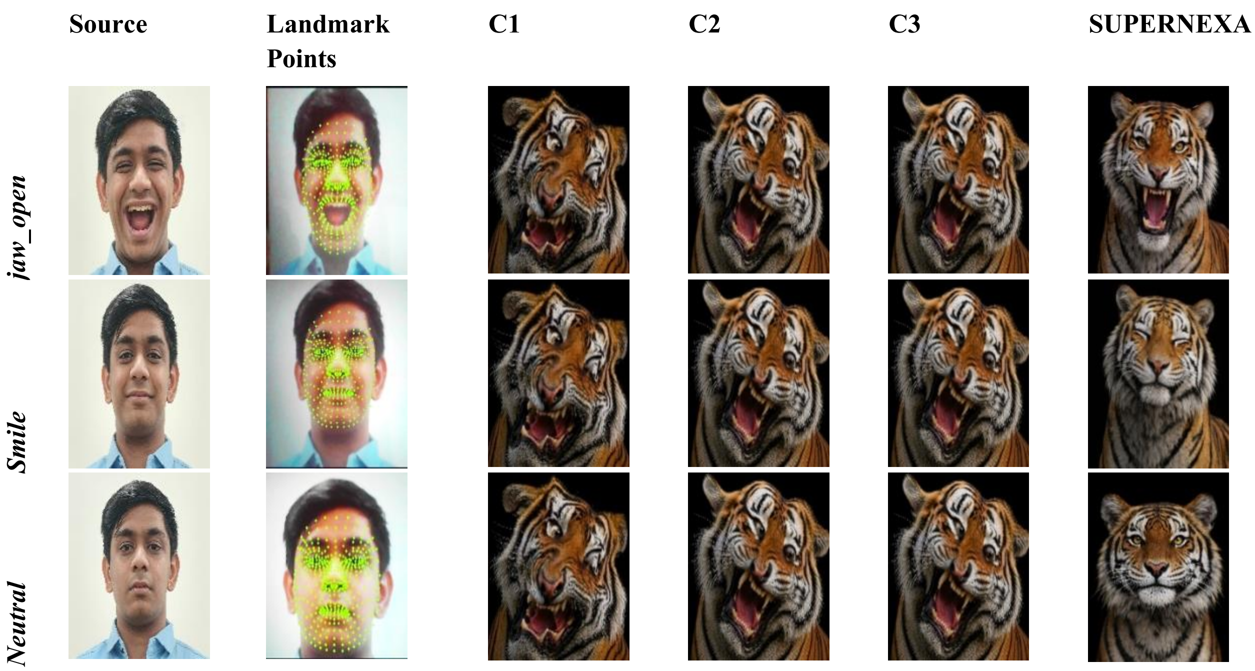

Fig. 4. Landmark-guided facial retargeting across configurations. Left to right: source, landmarks, C1, C2, C3, and SUPERNEXA outputs. Rows denote different expressions.

6.4.3 Visual Evaluation of Facial Retargeting Across Diverse Character Domains

Fig. 5 demonstrates the generalization ability of SUPERNEXA in terms of different target characters (Na'vi, Toruk, Cyberpunk, and Tiger). SUPERNEXA is effective in transferring semantic expressions regardless of changes in facial geometry, texture, and topology. It effectively transfers high-displacement expressions such as jaw_open without any distortion in its deformation. It is equally effective when it comes to transfer subtle expressions such as smile and eye_blink in different characters.

Fig. 5. Expression transfer across different character domains. Left to right: source, Na'vi, Toruk, Cyberpunk, and Tiger outputs. Rows denote expressions.

In summary, the results from the qualitative analysis are in line with those achieved through the quantitative approach, illustrating the superior performance of the SUPERNEXA model. Irrespective of the various expressions used, along with the choice of the target characters, the model is able to generate precise facial motion animations by accurately localizing spatially and ensuring seamless temporal animation. It is evident that the fusion of TPS and GRU methods is critical for attaining quality facial animation.

Conclusion:

Scalable Unified Performance Retargeting via Neural EXpressive Animation a real time facial retargeting (SUPERNEXA) system is introduced to transfer human expressions seamlessly to heterogenous digital characters. The proposed system combines geometric landmark representation, GRU based temporal refinement, TPS based deformation, and gaze retargeting into one unified framework. The experimental results of proposed framework demonstrate that the proposed architecture, successfully achieved accurate, temporally stable and noticeably realistic facial animation. The combination of TPS based spatial deformation and GRU based smoothing lead to significant improvement in the performance of retargeting in proposed architecture. The proposed architecture demonstrated that, it reduced Landmark error from 0.0269 to 0.0078, showcasing approximately 71.0% improvement, while the Motion Jitter reduced from 0.0736 to 0.0226, achieving nearly 69.3% improvement in temporal stability. Additionally, the Deformation Accuracy increased from 0.9526 to 0.9853 while improving the spatial Coverage from 0.445 to 0.467 during expression transfer. Furthermore, the proposed architecture effectively preserved expression consistency across different character topologies including Na'vi, Tiger, Cyberpunk, and Toruk characters while maintaining real time GRU based performance which was approximately 25-40 FPS. The obtained results validate the effectiveness of this framework in generating anatomically, visually stable and consistently expressive facial animations without needing predefined rigs or training datasets. Scalable Unified Performance Retargeting via Neural EXpressive Animation therefore provides a scalable and effective foundation for future facial animation systems, gaming, film production and virtual avatars.

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